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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: SOA PROGRAMMING AND MICROSERVICES– (24SDCS03)
A.Y. 2026 - 2027

TITLE OF THE PROJECT	AI-Driven Integrated Course Registration and Student Academic Management System for Intelligent Academic Planning and Administration	
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Abstract:

University course registration and academic administration traditionally rely on disjointed, manually operated systems that handle enrollment, timetable clashes, prerequisite checks, and academic advising as separate, loosely connected activities. This fragmentation often results in registration bottlenecks, seat-allocation conflicts, delayed academic guidance, and administrative overhead for both students and faculty. As student populations and elective course catalogues grow, the need for a unified, intelligent, and scalable platform that automates registration while proactively supporting academic planning has become increasingly critical.

To address this need, this project proposes an AI-Driven Integrated Course Registration and Student Academic Management System designed for intelligent academic planning and administration. The system unifies course registration, student records management, and academic advising into a single cohesive platform, augmented with AI-based recommendations to help students plan course loads, detect scheduling conflicts, and track progress toward degree requirements in real time.

The system is built on a Service-Oriented Architecture (SOA) using Spring Boot microservices, with each core function such as course registration, student record management, and academic planning implemented as an independently deployable service that communicates through REST APIs. This modular design improves scalability, fault isolation, and maintainability, allowing individual services to be updated or scaled without disrupting the rest of the system. A React.js single-page application delivers a responsive front-end experience, while JWT-based authentication and authorization secure every API endpoint and enforce role-based access for students, faculty, and administrators.

Intelligent academic planning is powered by the Gemini API, which is integrated into the planning service to generate personalized course recommendations, flag potential scheduling conflicts, and provide natural-language guidance based on a student's academic history and remaining degree requirements. All persistent data, including course catalogues, enrollment records, and student academic history, is managed through MySQL databases, with each microservice maintaining its own schema to preserve data ownership and consistency across the distributed system.

The development workflow follows industry-standard engineering practices: Maven manages dependency resolution and builds for the Spring Boot services, Postman is used to design, test, and validate REST API contracts during development, and Git/GitHub supports version control and collaborative development across the team. Together, these tools and technologies form a robust, end-to-end pipeline from API design through deployment.

Collectively, this system aims to reduce administrative overhead, minimize registration conflicts, and give students clearer, AI-assisted visibility into their academic journey. By combining a microservices-based backend, a modern React front-end, secure JWT authentication, and AI-driven planning through the Gemini API, the proposed platform offers a scalable and intelligent foundation for course registration and academic management in higher education institutions.

Project Architecture/Project Overview:

AI-Driven Integrated Course Registration & Student Academic Management System

Layered Microservices Architecture

